

Adaptive Intent-Preference Arbitration for Conversational Recommendation

Dhayanidhi Rajasekaran and Rajalakshmi N.R.

Abstract—A conversational recommender hears two voices at once: what a user has liked over many sessions, and what they are asking for in the current conversation. Most systems blend the two with a fixed rule and hope for the best. That works when the signals agree and fails quietly when they do not, for example when a habitual comedy viewer asks for a war film for the weekend. We propose *Adaptive Intent-Preference Arbitration* (AIPA), a model that first classifies the relationship between the long-term preference (LTP) and the short-term intent (STI) into one of five categories (complement, consistent, conflict, override, uncertain), then chooses an action (fuse, prioritise LTP, prioritise STI, or ask a clarifying question), and finally re-ranks candidates with learned, action-conditioned fusion weights. Two further components track whether an intent shift repeats across sessions and should be absorbed into the long-term profile, and probe the trained model with counterfactual ablations of each signal to report which one actually drove a recommendation. We build the whole pipeline on ReDial, an English human-human movie recommendation corpus, deriving cross-session profiles from recurring seeker identities and using MovieLens genres as item metadata. Because ReDial has no relationship annotations, labels come from a transparent weak-rule scheme and from controlled synthetic injection of all five relationship classes, and every result is reported separately for the two sources. Training emphasises disagreement turns through conflict-weighted losses, relationship label smoothing and confidence-filtered self-training, with every choice made on the validation split. In the reduced-budget run reported here (25% of the data, 2 seeds) AIPA is clearly better than profile-only, GRU and SASRec baselines and on par with a KBRD-style conversational baseline, but it remains slightly below an intent-only model on natural turns, does not separate from any fusion baseline on conflict turns, and the ablations are statistically flat. The arbitration head itself behaves sensibly: clarification precision is 1.0 on natural turns and wrong overrides are below 2%. We report these outcomes as they are, discuss why the central hypothesis is not yet supported, and list the concrete steps needed for a conclusive test. All numbers, tables and figures in this paper are generated by the released code.

Index Terms—Conversational recommender systems, user intent, long-term preference, preference conflict, clarification, counterfactual analysis, reproducibility.

I. INTRODUCTION

RECOMMENDER systems that talk to their users have an advantage that click-based systems lack: the user can simply say what they want right now. The flip side is that this fresh signal has to be reconciled with everything the system

already knows about the person. Someone who has spent years watching dramas may ask, on a Friday night, for “something dumb and funny”. A system that follows its profile ignores the request; a system that follows the request may throw away a decade of evidence for a single evening. Neither extreme is right, and the right answer depends on the situation.

The bulk of work on conversational recommendation (CRS) has focused on how to elicit preferences within a dialogue [1]–[4] or how to ground dialogue in knowledge about items [5], [6]. How to reconcile what is said now with what was learned before has received much less direct attention. When both signals are used, they are usually combined with a fixed rule or a learned gate that is never asked to explain itself, and evaluation almost never singles out the turns where the two disagree. Yet those are exactly the turns where the design choice matters.

This paper treats the reconciliation step as a first-class problem. We call it *arbitration*, and we make it explicit in three ways. First, the model predicts what kind of relationship holds between the long-term preference (LTP) and the short-term intent (STI): they may reinforce each other, one may add a new dimension the other lacks, they may contradict each other, the user may explicitly set the profile aside, or there may simply be too little evidence to tell. Second, this relationship drives a discrete action (fuse the signals, favour one of them, or ask a question) and the action in turn sets the fusion weights. Third, the model is made to account for its recommendations: after training we knock out one signal at a time and measure how much the ranking moves, giving a per-turn label of which signal drove the recommendation. We stress that this is a diagnostic of the fitted model, not a causal claim about users.

Our second aim is a clean and fully reproducible experimental base. We deliberately use ReDial [7], an English corpus of human-human movie dialogues. It has no cross-session profiles by design, but crowd workers recur across many dialogues, and we use that recurrence to construct chronological seeker histories under strict no-look-ahead rules. ReDial also has no labels for our relationship taxonomy; rather than pretend otherwise, we derive weak-rule labels from the dialogue text and additionally inject controlled synthetic conflicts, and we never mix the two in a table. Every claim about a metric is accompanied by bootstrap intervals and paired tests with multiple-comparison correction.

The results of the run reported here are mixed, and we say so. AIPA improves on profile-only and sequential baselines (GRU, SASRec) by a clear margin, matches a KBRD-style conversational baseline, its arbitration behaviour is sensible,

D. Rajasekaran and N.R. Rajalakshmi are with the Department of Computer Science and Engineering, Vel Tech Rangarajan Dr Sagunthala R&D Institute of Science and Technology, Chennai, India. E-mail: dhayanidhi.r@gmail.com, rajalakshmi234@gmail.com.

Code, configuration, generated figures and the executed notebook are available at <https://github.com/dhayanidhi/AIPA-CRS>.

and on controlled conflicts it follows the stated intent. It does not, however, outperform an intent-only recommender, which is in fact slightly and significantly better on the natural test set, and none of the ablations reach significance. We see the contribution of this paper as the framing, the evaluation protocol, and an honest baseline measurement that later work can build on; the discussion in Section VIII lists what a decisive test would need.

II. RELATED WORK

A. Conversational recommendation

Early CRS work framed the dialogue as a preference elicitation game in which the system asks about attributes and the user answers [1], [2]. Estimation–Action–Reflection [3] formalised the choice between asking and recommending as a policy problem. The “system ask, user respond” paradigm of Zhang *et al.* [4] likewise treats clarifying questions as an action to be planned. Our clarification action sits in this tradition but is triggered by the relationship between two evidence sources rather than by attribute uncertainty alone.

Open-ended, natural-language CRS became tractable with ReDial [7], followed by knowledge-grounded models such as KBRD [5] and KGSF [6], and by further corpora including INSPIRED [8] and the Chinese TG-ReDial [9]. TG-ReDial is the only one of these that ships per-user histories, but its dialogues are in Chinese; we chose ReDial so that data, labels, case studies and error analysis could all be read in English, and reconstructed histories from recurring seekers instead. Surveys by Jannach *et al.* [10] and Gao *et al.* [11] give a broader map of the field; both note that reconciling the conversation with the historical profile remains open.

B. Long- and short-term signals

Session-based and sequence-aware recommenders [12]–[14] learn to weigh recent behaviour against older behaviour implicitly, through the recurrence or the attention pattern. Our sequential baseline follows this recipe. The difference in AIPA is that the weighing is exposed: the relationship class and the action are discrete, supervised (weakly) and inspectable, and the fusion weights are a function of them.

C. Counterfactual reasoning in recommendation

We borrow the vocabulary of interventions [15] for the driver diagnostic, but the object we intervene on is the trained model, not the user. Zeroing the LTP encoding and re-scoring tells us how much the model’s output depends on that pathway, which is a form of ablation-based attribution. We do not claim identification of any causal effect on user behaviour, and we avoid the word “effect” when describing these numbers.

III. PROBLEM FORMULATION

This section fixes notation and states the prediction problem exactly as the released code constructs it. Table I collects the symbols. Vectors are bold lower case ($\mathbf{h}$), matrices bold upper case ($\mathbf{W}$), sets calligraphic ($\mathcal{I}$), and $\mathbb{I}[\cdot]$ is the indicator function. For a vector $\mathbf{v}$ with non-negative entries we write

TABLE I
NOTATION.

Symbol	Meaning
$\mathcal{I}, \mathcal{G}$	item catalogue; genre set ($ \mathcal{G} = 19$)
$G(i) \subseteq \mathcal{G}$	MovieLens genres of item i (may be empty)
u, c, t	seeker, dialogue (conversation id), turn index
$\mathcal{H}_u(c)$	dialogues of u strictly before c , Eq. (1)
$\mathbf{h} = (h_1, \dots, h_L)$	liked/seen items from $\mathcal{H}_u(c)$, $L \leq L_{\max}$
$\mathcal{P}$	profile sentences of u from $\mathcal{H}_u(c)$
R_t, ℓ_t	recent seeker text and last seeker utterance before t
$\mathcal{M}_{<t}, \mathcal{M}_{<t}^+$	movies mentioned / liked by the seeker before t
$\pi_L, \pi_S \in \bar{\Delta}^{ \mathcal{G} }$	LTP and STI genre vectors, Eqs. (3), (4)
$\mathbf{f} \in \mathbb{R}^8$	lexical and length flags, Eq. (5)
$\phi(\cdot) \in \mathbb{R}^{d_t}$	text encoder, Eq. (6)
$\mathbf{e}_i, b_i$	item vector and item bias, Eq. (7)
$\mathbf{h}_L, \mathbf{h}_S \in \mathbb{R}^d$	LTP and STI encodings
$\mathbf{s}_L, \mathbf{s}_S, \mathbf{s} \in \mathbb{R}^{ \mathcal{I} }$	component and fused score vectors
$\mathcal{R}, \mathcal{A}$	relationship classes, actions ($ \mathcal{R} = 5, \mathcal{A} = 4$)
$\mathbf{p} \in \Delta^5, \mathbf{q} \in \Delta^4$	relationship posterior, action distribution
$\mathbf{W}_A \in \mathbb{R}^{4 \times 2}$	action-to-weight table (learned), Eq. (17)
$\mathbf{c} \in \mathbb{R}^5$	counterfactual features, Eq. (23)
$\gamma, \theta, \tau, \rho, k, \eta$	decay, rule threshold, driver threshold, dominance ratio, persistence count, persistence gain

$\text{nrm}(\mathbf{v}) = \mathbf{v} / \mathbf{1}^\top \mathbf{v}$ if $\mathbf{1}^\top \mathbf{v} > 0$ and $\text{nrm}(\mathbf{0}) = \mathbf{0}$, so that genre vectors live in $\bar{\Delta}^{|\mathcal{G}|} = \Delta^{|\mathcal{G}|} \cup \{\mathbf{0}\}$, the probability simplex extended by the “no evidence” point.

A. Seekers, sessions and the temporal boundary

ReDial is a set of dialogues, each a message sequence $D = (m_1, \dots, m_T)$ in which every message carries a role (seeker or recommender), English text and the set $M(m) \subseteq \mathcal{I}$ of movies it mentions. Every dialogue has a seeker identifier u and an integer conversation identifier c , and the seeker annotates each mentioned movie as liked, seen and suggested. We write $\mathcal{L}(D) \subseteq \mathcal{I}$ for the movies the seeker marked as liked or seen in D .

Conversation identifiers were assigned sequentially during collection, and we treat them as the chronological order of a seeker’s sessions (an assumption; see Section VIII). The history available at dialogue c of seeker u is then

$$\mathcal{H}_u(c) = \{D_{c'}^u : c' < c\}, \quad (1)$$

the set of u ’s dialogues with a strictly smaller identifier. Nothing from D_c^u itself or from any later dialogue enters the long-term side.

B. Instances

A recommendation instance is a triple (c, t, y) : at turn $t > 1$ of dialogue c the recommender’s message mentions a movie $y \in \mathcal{I}$ that has not appeared earlier in the dialogue,

$$y \in M(m_t) \setminus \bigcup_{t' < t} M(m_{t'}), \quad \text{role}(m_t) = \text{recommender}. \quad (2)$$

A recommender message introducing several new movies yields one instance per movie. The prediction problem is to rank the full catalogue $\mathcal{I}$ so that y is placed as high as possible, using only the two evidence bundles below.

a) *Long-term preference (LTP)*: From $\mathcal{H}_u(c)$ in chronological order we take the liked/seen items, keep the most recent $L_{\max} = 50$ of them, and call the result $\mathbf{h} = (h_1, \dots, h_L)$ with h_L the most recent. We also keep the last ten seeker utterances in $\mathcal{H}_u(c)$ that match a preference-statement pattern (“I usually like...”, “my favourite...”) and are shorter than 200 characters; their concatenation is the profile text $\mathcal{P}$. The LTP genre vector is a recency-decayed mixture of the items’ genre indicators,

$$\pi_L = \text{nrm}\left(\sum_{j=1}^L \gamma^{L-j} \frac{\mathbf{g}(h_j)}{|G(h_j)|}\right), \quad \mathbf{g}(i)_g = \mathbb{1}[g \in G(i)], \quad (3)$$

with $\gamma = 0.9$, so the most recent item has weight one and each older item is discounted by a further factor γ . Items with no MovieLens match ($G(i) = \emptyset$) contribute nothing. A seeker with $L < L_{\min} = 3$ history items is *cold*.

b) *Short-term intent (STI)*: Let $S_{<t}$ be the seeker messages before turn t , R_t the concatenated text of the last 6 of them, and ℓ_t the text of the last one. Let $\mathcal{M}_{<t}$ be the movies mentioned in $S_{<t}$ and $\mathcal{M}_{<t}^+ \subseteq \mathcal{M}_{<t}$ those the seeker annotated as liked. Titles in the text are substituted by their surface form, so “@12345” becomes the movie name. The STI genre vector combines lexicon counts with the genres of liked in-dialogue movies,

$$\pi_S = \text{nrm}\left(\mathbf{n}(R_t) + \text{nrm}\left(\sum_{i \in \mathcal{M}_{<t}^+} \frac{\mathbf{g}(i)}{|G(i)|}\right)\right), \quad (4)$$

where $\mathbf{n}(R_t)_g$ counts occurrences in R_t of the keywords of a small English lexicon for genre g (“scary”, “slasher”, “zombie” for Horror, and so on). Eight scalar flags summarise lexical and length evidence,

$$\mathbf{f} = (f_{\text{sti}}, f_{\text{tp}}, f_{\text{neg}}, f_{\text{req}}, \frac{|S_{<t}|}{10}, \frac{|\mathcal{M}_{<t}|}{5}, f_{\text{cold}}, \frac{L}{L_{\max}}), \quad (5)$$

where f_{sti} and f_{tp} indicate an explicit departure marker (“for a change”, “tonight”, “with my kids”) or an explicit habit marker (“as usual”, “stick with”) in R_t , f_{neg} indicates a negated genre keyword (“not into horror”), f_{req} indicates a request pattern in ℓ_t , and $f_{\text{cold}} = \mathbb{1}[L < L_{\min}]$.

Everything on the STI side is computed from $S_{<t}$ only; everything on the LTP side from $\mathcal{H}_u(c)$ only. The model therefore sees neither the target turn nor any later data.

C. Relationships, actions and shifts

We distinguish five relationship classes $\mathcal{R} = \{\text{Complement}, \text{Consistent}, \text{Conflict}, \text{Override}, \text{Uncertain}\}$ between the two bundles. *Consistent*: STI points where LTP already points. *Complement*: STI adds a genre that LTP does not cover without contradicting it. *Conflict*: STI opposes LTP with no sign that the user is aware of it. *Override*: STI opposes LTP and the user flags the departure explicitly. *Uncertain*: one side carries too little evidence. The action set is $\mathcal{A} = \{\text{Fuse}, \text{Prioritise_LTP}, \text{Prioritise_STI}, \text{Ask_Clarification}\}$, and a fixed default map $A_0 : \mathcal{R} \rightarrow \mathcal{A}$ sends Consistent and Complement to Fuse, Conflict and Override to Prioritise_STI, and Uncertain to Ask_Clarification. A learned policy may depart from A_0 .

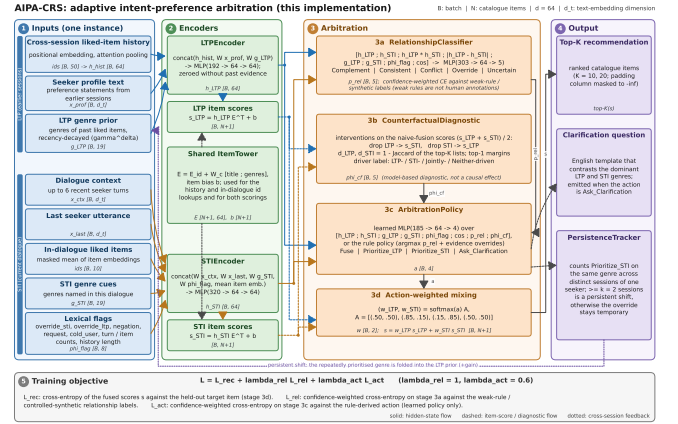


Fig. 1. AIPA components. History and dialogue context feed separate LTP and STI encoders. A relationship classifier and a counterfactual driver probe inform the arbitration policy, which produces fusion weights (or a clarifying question). A persistence tracker decides whether repeated intent shifts should update the long-term profile.

A departure from habit that occurs in one session is a *temporary override*. If the same departure wins arbitration in at least k distinct sessions of the same seeker it is a *persistent shift* and should be folded into LTP (Section IV-H).

IV. METHOD

Figure 1 shows the components. All encoders are small because the point of the work is the arbitration logic, not representational power; the item tower, the dimensionality $d = 64$ and the training recipe are shared by every compared system so that differences are attributable to the arbitration design. Throughout, $\text{MLP}(\mathbf{z}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2$ with hidden width d and dropout 0.1 after the activation; each occurrence has its own parameters.

A. Text and item representations

Text is encoded by a fixed (non-neural) map fitted on the training split: word uni- and bigram TF-IDF with sublinear term frequency, truncated SVD to $d_t = 64$ dimensions, and ℓ_2 normalisation,

$$\phi(x) = \frac{\mathbf{V}^\top \text{tfidf}(x)}{\|\mathbf{V}^\top \text{tfidf}(x)\|_2 + 10^{-8}}, \quad \phi(\text{empty}) = \mathbf{0}. \quad (6)$$

Item i has a content vector $\mathbf{x}_i = [\phi(\text{title}_i); \mathbf{g}(i)] \in \mathbb{R}^{d_t + |\mathcal{G}|}$ and the item tower produces

$$\mathbf{e}_i = \mathbf{E}_i + \mathbf{W}_c \mathbf{x}_i + \mathbf{b}_c, \quad \mathbf{b}_i \in \mathbb{R}, \quad (7)$$

with a free embedding table $\mathbf{E} \in \mathbb{R}^{|\mathcal{I}| \times d}$, a shared projection $\mathbf{W}_c$ and a free per-item bias $\mathbf{b}_i$. Given any query vector $\mathbf{h} \in \mathbb{R}^d$ the score of item i is the biased inner product

$$s_i(\mathbf{h}) = \mathbf{h}^\top \mathbf{e}_i + b_i, \quad (8)$$

and $\mathbf{s}(\mathbf{h}) \in \mathbb{R}^{|\mathcal{I}|}$ collects the scores of the whole catalogue (the padding index is masked to $-\infty$).

B. LTP encoder

The history $\mathbf{h}$ is right-aligned in a window of length $L_{\max}$ and each position j receives a learned positional vector $\mathbf{u}_j$. The encoder pools the item vectors with a single learned attention query $\mathbf{a} \in \mathbb{R}^d$,

$$\tilde{\mathbf{e}}_j = \mathbf{e}_{h_j} + \mathbf{u}_j, \quad \alpha_j = \frac{\exp(\mathbf{a}^\top \tilde{\mathbf{e}}_j)}{\sum_{j'=1}^L \exp(\mathbf{a}^\top \tilde{\mathbf{e}}_{j'})}, \quad (9)$$

$$\mathbf{h}_{\text{hist}} = \sum_{j=1}^L \alpha_j \tilde{\mathbf{e}}_j, \quad (10)$$

where the softmax runs over the non-padding positions only and $\mathbf{h}_{\text{hist}} = \mathbf{0}$ when $L = 0$. The positional vectors let the attention prefer recent or old items, but the recency discount itself is applied to the genre vector through γ in Eq. (3) rather than to the attention weights. The encoding is

$$\mathbf{h}_L = \mathbb{I}[\text{LTP evidence}] \cdot \text{MLP}([\mathbf{h}_{\text{hist}}; \mathbf{W}_p \phi(\mathcal{P}); \mathbf{W}_g \pi_L]), \quad (11)$$

where the indicator is one if at least one of $\mathbf{h}$, π_L or $\phi(\mathcal{P})$ is non-zero. A seeker without any history thus has $\mathbf{h}_L = \mathbf{0}$ and, by Eq. (8), an LTP score vector equal to the item biases.

C. STI encoder

The intent side has no sequence to attend over; it concatenates five projected views of the current dialogue, one of them a masked mean of the liked in-dialogue items (at most the last ten),

$$\bar{\mathbf{e}}_{\text{cur}} = \frac{1}{\max(1, |\mathcal{M}_{<t}^+|)} \sum_{i \in \mathcal{M}_{<t}^+} \mathbf{e}_i, \quad (12)$$

$$\mathbf{h}_S = \text{MLP}([\mathbf{W}_r \phi(R_t); \mathbf{W}_\ell \phi(\ell_t); \mathbf{W}_g' \pi_S; \mathbf{W}_f \mathbf{f}; \bar{\mathbf{e}}_{\text{cur}}]). \quad (13)$$

D. Scoring and the fusion family

Both encodings are scored against the catalogue with Eq. (8), $\mathbf{s}_L = \mathbf{s}(\mathbf{h}_L)$ and $\mathbf{s}_S = \mathbf{s}(\mathbf{h}_S)$, and every compared system produces its final ranking from a convex combination

$$\mathbf{s} = w_L \mathbf{s}_L + w_S \mathbf{s}_S, \quad w_L + w_S = 1, \quad w_L, w_S \geq 0. \quad (14)$$

Because the score is linear in the query, Eq. (14) equals $\mathbf{s}(w_L \mathbf{h}_L + w_S \mathbf{h}_S)$: mixing scores and mixing encodings are the same operation, and the item bias enters exactly once. The systems differ only in how (w_L, w_S) is chosen:

$$(w_L, w_S) = \begin{cases} (1, 0) & \text{LTP-only,} \\ (0, 1) & \text{STI-only,} \\ (\alpha, 1 - \alpha) & \text{naive fusion, } \alpha = 0.5, \\ (\sigma(z), 1 - \sigma(z)) & \text{adaptive fusion,} \\ \mathbf{q}^\top \mathbf{W}_A & \text{AIPA.} \end{cases} \quad (15)$$

The naive weight is also swept over $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ at test time (Table IX). Adaptive fusion learns a scalar gate

$$z = \text{MLP}([\mathbf{h}_L; \mathbf{h}_S; \pi_L; \pi_S; \mathbf{f}]) \in \mathbb{R}, \quad (16)$$

and has no notion of relationship or action. AIPA replaces the gate by a distribution $\mathbf{q} \in \Delta^4$ over actions (Section IV-G) and

a table that states what each action means for the weights. Each row of the table is the softmax of two free logits $\ell_a \in \mathbb{R}^2$,

$$[\mathbf{W}_A]_{a \cdot} = \text{softmax}(\ell_a), \quad \mathbf{W}_A^{(0)} = \begin{pmatrix} 0.50 & 0.50 \\ 0.85 & 0.15 \\ 0.15 & 0.85 \\ 0.50 & 0.50 \end{pmatrix} \begin{matrix} \text{Fuse} \\ \text{Prioritise_LTP} \\ \text{Prioritise_STI} \\ \text{Ask_Clarification} \end{matrix} \quad (17)$$

so that $w_L = \sum_a q_a [\mathbf{W}_A]_{a1}$ and the constraint in Eq. (14) holds automatically. The logits are initialised so that $\mathbf{W}_A$ equals the hand-set table $\mathbf{W}_A^{(0)}$ and are then trained end-to-end through the recommendation loss (learned_action_weights); the previous fixed-table behaviour is retained as a configuration option. The meaning of each action therefore remains inspectable (one weight pair per action), but the pair itself is fitted; in the run reported here the Prioritise_STI row moved to roughly 0.22/0.78 and the Fuse row stayed near 0.5/0.5 (Table XI). An optional instance-level residual $g \in \mathbb{R}$ from the policy, $w_L \leftarrow \sigma(\text{logit}(w_L) + g)$, is implemented but switched off in every reported configuration. A clarifying turn still ranks the catalogue with the Ask_Clarification row, so an offline ranking metric is defined for every instance.

E. Relationship classifier

The classifier sees both encodings, their element-wise interactions, the raw genre vectors and flags, and the cosine of the encodings,

$$\boldsymbol{\varphi}_{\text{rel}} = [\mathbf{h}_L; \mathbf{h}_S; \mathbf{h}_L \odot \mathbf{h}_S; |\mathbf{h}_L - \mathbf{h}_S|; \pi_L; \pi_S; \mathbf{f}; \cos(\mathbf{h}_L, \mathbf{h}_S)], \quad (18)$$

$$\mathbf{r} = \text{MLP}(\boldsymbol{\varphi}_{\text{rel}}) \in \mathbb{R}^5, \quad \mathbf{p} = \text{softmax}(\mathbf{r}), \quad (19)$$

with $\cos(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top \mathbf{y} / \max(\|\mathbf{x}\| \|\mathbf{y}\|, 10^{-6})$. The posterior $\mathbf{p}$ is passed to the policy through a stop-gradient, so the action loss cannot reshape the relationship head; the relationship head is trained only by its own (weak) labels.

F. Counterfactual driver diagnostic

The diagnostic asks how much a ranking depends on each pathway, and answers by intervening on the model rather than on the data. The reference ranking is the naive fusion of the two component scores, and removing a pathway is defined as scoring with the other pathway alone,

$$\mathbf{s}_\emptyset = \frac{1}{2}(\mathbf{s}_L + \mathbf{s}_S), \quad \mathbf{s}_\emptyset|_{do(L=\emptyset)} := \mathbf{s}_S, \quad \mathbf{s}_\emptyset|_{do(S=\emptyset)} := \mathbf{s}_L. \quad (20)$$

This is a convention rather than a literal $\mathbf{h}_L \leftarrow \mathbf{0}$ (which would give $\frac{1}{2}\mathbf{h}_S^\top \mathbf{e}_i + b_i$ and weigh the item bias twice as heavily); it keeps the interventional list equal to what the STI or LTP tower would recommend on its own. Disruption is one minus the Jaccard overlap of the factual and interventional top- K lists ($K = 10$),

$$J_K(\mathbf{s}, \mathbf{s}') = \frac{|\text{Top}_K(\mathbf{s}) \cap \text{Top}_K(\mathbf{s}')|}{|\text{Top}_K(\mathbf{s}) \cup \text{Top}_K(\mathbf{s}')|}, \quad (21)$$

$$\delta_L = 1 - J_K(\mathbf{s}_\emptyset, \mathbf{s}_S), \quad \delta_S = 1 - J_K(\mathbf{s}_\emptyset, \mathbf{s}_L), \quad (22)$$

so δ_L measures how much the list moves when LTP is removed and δ_S when STI is removed. Together with the top-1 margins $m(\mathbf{s}) = s_{(1)} - s_{(2)}$ of the three score vectors they form the feature vector handed to the policy,

$$\mathbf{c} = (\delta_L, \delta_S, m(\mathbf{s}_\emptyset), m(\mathbf{s}_S), m(\mathbf{s}_L)) \in \mathbb{R}^5, \quad (23)$$

computed without gradient. The per-turn driver label uses a threshold $\tau = 0.1$ and a dominance ratio $\rho = 1.5$:

$$\text{driver} = \begin{cases} \text{Neither} & \delta_L < \tau \wedge \delta_S < \tau, \\ \text{STI} & \delta_S \geq \tau \wedge \delta_S \geq \rho \delta_L, \\ \text{LTP} & \delta_L \geq \tau \wedge \delta_L \geq \rho \delta_S, \\ \text{Jointly} & \text{otherwise,} \end{cases} \quad (24)$$

with the cases tested in the order written. The labels describe the sensitivity of the fitted model; nothing here identifies an effect on users. A second, post-training probe re-scores the *fused* model of Eq. (14) under $\mathbf{h}_L \leftarrow \mathbf{0}$ and $\mathbf{h}_S \leftarrow \mathbf{0}$ (which also changes $\mathbf{p}$, $\mathbf{q}$ and hence the weights) and reports rank and NDCG changes and the plain overlap $|\text{Top}_{10} \cap \text{Top}'_{10}|/10$; that is the quantity in Table VIII.

G. Arbitration policy and clarification trigger

a) *Learned policy*: The action logits are a function of the encodings, the evidence, the (detached) relationship posterior and the counterfactual features,

$$\varphi_{\text{act}} = [\mathbf{h}_L; \mathbf{h}_S; \pi_L; \pi_S; \mathbf{f}; \cos(\mathbf{h}_L, \mathbf{h}_S); \text{sg}(\mathbf{p}); \mathbf{c}], \quad (25)$$

$$\mathbf{o} = \text{MLP}(\varphi_{\text{act}}) \in \mathbb{R}^4, \quad \mathbf{q} = \text{softmax}(\mathbf{o}), \quad (26)$$

where sg is the stop-gradient. The ablations remove $\text{sg}(\mathbf{p})$ or $\mathbf{c}$ from Eq. (25); the “without clarification” ablation sets $\mathbf{o}_{\text{Ask}} = -\infty$ before the softmax. The weights follow from Eq. (17) and the emitted action is $\hat{a} = \arg \max_a q_a$.

b) *Rule policy*: The rule variant replaces Eq. (26) by a deterministic function of the relationship posterior and two evidence tests. With $\hat{r} = \arg \max_r p_r$, confidence $\kappa = \max_r p_r$, threshold $\theta = 0.5$, cold = $\mathbb{I}[f_{\text{cold}} > \frac{1}{2}]$ and noSTI = $\mathbb{I}[\pi_S = \mathbf{0}]$,

$$\hat{a} = \begin{cases} \text{Prioritise_STI} & \text{cold} \wedge \neg \text{noSTI}, \\ \text{Ask_Clarification} & \kappa < \theta, \\ A_0(\hat{r}) & \text{otherwise,} \end{cases} \quad (27)$$

where the first case takes precedence. When clarification is disabled the second case is dropped, any Ask_Clarification produced by A_0 becomes Fuse, and a non-cold seeker with noSTI is sent to Prioritise_LTP. The rule output is turned into logits $\mathbf{o} = 20 \cdot \mathbf{1}_{\hat{a}}$, so its softmax is a near one-hot $\mathbf{q}$ and Eq. (17) returns the weight row of $\hat{a}$ almost exactly.

c) *Clarification*: A clarifying question is emitted whenever $\hat{a} = \text{Ask_Clarification}$: for the learned policy this is the arg-max of Eq. (26), for the rule policy the second case of Eq. (27). The question is filled from a small English template set using the arg-max genres of π_L and π_S , for example “Earlier you enjoyed drama movies, but right now you seem interested in horror. Should I focus on horror for tonight, or find something that fits both?” Offline we cannot observe the answer, so the question is evaluated by whether the reference label called for it (Section VI-B).

H. Persistence tracker

At inference the natural test instances are visited in order of (seeker, conversation id, turn). For each seeker u and genre g the tracker keeps the set $\mathcal{C}_{u,g}$ of sessions in which the first instance of the session was resolved by Prioritise_STI with g the arg-max of π_S ,

$$\mathcal{C}_{u,g} \leftarrow \mathcal{C}_{u,g} \cup \{c\} \quad \text{if } \hat{a} = \text{Prioritise_STI} \wedge g = \arg \max_{g'} \pi_{S,g'}. \quad (28)$$

Before an instance of seeker u is scored, its LTP genre vector is replaced by

$$\pi'_L = \text{nrm} \left(\pi_L + \eta \sum_{g: |\mathcal{C}_{u,g}| \geq k} \mathbf{1}_g \right), \quad (29)$$

with $k = 2$ and $\eta = 0.3$, where $\mathbf{1}_g$ is the unit vector of genre g ; only sessions already visited count, so the adjustment never uses the future. Each time $|\mathcal{C}_{u,g}|$ first reaches k a persistent shift (u, g, c) is logged. The model is then re-run with π'_L in place of π_L ; the encoders and the policy are unchanged, so the tracker acts only through Eqs. (11), (18), (16) and (25). Synthetic instances are excluded from the tracker.

I. Training objective

Every instance n carries a target y_n and, for AIPA variants, a relationship label $r_n \in \mathcal{R}$, an action label $a_n \in \mathcal{A}$ and a confidence $\omega_n \in [0, 1]$ from the labelling procedure of Section V ($\omega_n = 1$ for synthetic instances). With $\text{CE}(\mathbf{z}, y) = -\log \text{softmax}(\mathbf{z})_y$ and CE_ϵ its label-smoothed version, the mini-batch loss is

$$\mathcal{L} = \frac{1}{B} \sum_{n=1}^B \left[\nu_n \mathcal{L}_{\text{rec}}(\mathbf{s}_n, y_n) + \omega_n (\lambda_{\text{rel}} \text{CE}_\epsilon(\mathbf{r}_n, r_n) + \lambda_{\text{act}} \text{CE}(\mathbf{o}_n, a_n)) \right], \quad (30)$$

with $\lambda_{\text{rel}} = 1.0$, $\lambda_{\text{act}} = 0.6$ and $\epsilon = 0.1$. Three ingredients target disagreement turns. *Conflict weighting*. $\nu_n = 2.0$ for instances whose label is Conflict or Override (natural weak-rule or synthetic) and 1 otherwise, renormalised to mean one within the batch so that the overall loss scale is unchanged (conflict_loss_weight; 1 recovers the unweighted loss). *Recommendation loss*. $\mathcal{L}_{\text{rec}}$ is configurable as the full softmax over the catalogue, a sampled softmax with in-batch positives plus 256 uniform negatives, or BPR. The three were compared on the validation split in quick mode (validation Hit@10 of AIPA (full): full softmax 0.123, sampled softmax 0.072, BPR 0.046) and the full softmax is used in every reported run. *Self-training*. From epoch 3 on, a weak-rule relationship label with $\omega_n < 0.6$ is replaced by the model’s own prediction when its softmax maximum is at least 0.9; the decision is always taken against the original weak labels (so relabelling cannot drift), synthetic labels are never touched, and the action label follows A_0 of the new relationship. In the reported run this relabelled 44–112 of the 9,342 training instances per epoch. The two auxiliary terms are present only for AIPA variants, the action term only when the policy is learned, and each is scaled per instance by the label confidence so that a low-confidence weak label moves the heads less than a confident one. All of λ_{rel} , λ_{act} , the conflict weight, the smoothing and the self-training switch were selected on the validation split (natural validation Hit@10 as the primary

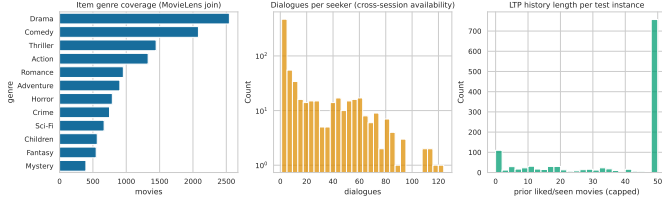


Fig. 2. ReDial overview: dialogue counts per split, dialogues per seeker, genre coverage after the MovieLens join, and turns per dialogue.

criterion, validation conflict-subset Hit@10 and macro-F1 reported alongside; grid $\lambda_{\text{rel}} \in \{0.5, 1\} \times \lambda_{\text{act}} \in \{0.1, 0.6\}$, all within 0.002 Hit@10 of each other; the test split was not used for any selection. For the “without clarification” ablation the action label Ask_Clarification is remapped to Fuse in Eq. (30) as well as being masked in Eq. (26), so the ablation is not merely a decoding constraint. All models are trained with AdamW [16] (learning rate 0.003, weight decay 10^{-5}), batch size 128, gradient-norm clipping at 5, for at most 15 epochs with early stopping on validation Hit@10 (patience three) and restoration of the best epoch. Baselines that are not fusion models (GRU, conversation-aware) use the same item tower, Eq. (8) and the first term of Eq. (30).

V. DATA AND LABELS

A. ReDial and cross-session histories

ReDial [7] contains 11,348 English dialogues (10,006 training, 1,342 test) between a seeker and a recommender, with per-movie seeker annotations for *liked*, *seen* and *suggested*. We split 1,000 training dialogues off as validation. The corpus has 764 distinct seeker identifiers and most seekers appear in many dialogues (Fig. 2). We order a seeker’s dialogues by conversation identifier and treat this as chronological order; this is an assumption we could not verify against timestamps, and we flag it as such. For dialogue D_{m+1} , LTP is built only from $D_1, \dots, D_m$; a seeker with fewer than 3 earlier sessions is treated as cold. Item metadata (genres, years) come from MovieLens *ml-latest* [17] joined on normalised title and year; movies without a match keep an empty genre vector rather than an imputed one.

Each recommendation instance is one (dialogue, turn, target movie) triple where the recommender introduces a movie not yet mentioned in the dialogue. Only the preceding 6 turns of context and at most 50 history items are used. In the run reported here we sample 25% of the dialogues in every split with a fixed seed, which yields 9342 training instances and 1077 natural test instances.

B. Relationship labels

ReDial has no relationship annotations. We use two label sources and keep them apart throughout.

Weak-rule labels. A deterministic function of $(\pi_L, \pi_S, R_t, \mathbf{f})$ assigns a relationship r_n , an action a_n and a confidence ω_n . Let $\text{sim} = \cos(\pi_L, \pi_S)$, let $T(\pi)$ be the (at most two) largest genres of π with mass at least 0.15, and let $\text{hasL} = \mathbb{K}[\pi_L \neq \mathbf{0} \wedge \neg f_{\text{cold}}]$. Three lexical tests read R_t : a habit marker (f_{itp}), a departure marker

(f_{sti}), and $\text{negH} = \mathbb{K}[\exists g \text{ negated in } R_t : \pi_{L,g} \geq 0.2]$. Two structural tests compare the genre vectors: $\text{tens} = \mathbb{K}[\exists s \in T(\pi_S), g \in T(\pi_L) : \{s, g\} \in \mathcal{T} \wedge \pi_{L,s} < 0.1]$ for a fixed list $\mathcal{T}$ of genre pairs that rarely co-occur in a request (Horror–Children, Comedy–War, ...), and $\text{new} = \mathbb{K}[\exists s \in T(\pi_S) : \pi_{L,s} < 0.05]$. Writing C/O for Override if $f_{\text{sti}} = 1$ and Conflict otherwise, the rule is evaluated top to bottom:

$$(r_n, a_n, \omega_n) = \begin{cases} (\text{Uncertain}, \text{Ask}, 0.9) & \pi_S = \mathbf{0} \wedge \neg \text{hasL}, \\ (\text{Uncertain}, \text{Ask}, 0.6) & \pi_S = \mathbf{0}, \\ (\text{Uncertain}, \text{Prio_STI}, 0.6) & \neg \text{hasL}, \\ (\text{Consistent}, \text{Fuse}, 0.8) & f_{\text{itp}} = 1, \\ (\text{C/O}, \text{Prio_STI}, 0.8) & \text{negH}, \\ (\text{C/O}, \text{Prio_STI}, \omega_t) & \text{tens} \wedge \text{sim} < 0.25, \\ (\text{Consistent}, \text{Fuse}, \omega_s) & \text{sim} \geq 0.5, \\ (\text{Complement}, \text{Fuse}, \omega_n) & \text{new}, \\ (\text{Consistent}, \text{Fuse}, 0.5) & \text{sim} \geq 0.25, \\ (\text{Uncertain}, \text{Ask}, 0.5) & \text{otherwise}, \end{cases} \quad (31)$$

with $\omega_s = 0.5 + 0.5 \text{sim}$, $\omega_n = 0.5 + 0.3(1 - \text{sim})$, $\omega_t = 0.75$ if $f_{\text{sti}} = 1$ and $\omega_t = 0.55 + 0.2(0.25 - \text{sim})/0.25$ otherwise. The action label follows A_0 except in the third case, where a cold seeker with a stated intent is labelled Prioritise_STI rather than Ask. These labels are noisy by construction; ω_n down-weights them in Eq. (30) and defines the “necessary clarification” event in Eq. (37).

Synthetic controlled labels. For 0.15 of natural test instances whose seeker has a usable LTP, we append one English seeker utterance and set the intended relationship. For *Conflict*, *Override* and *Consistent* the utterance requests a genre that conflicts with, overrides, or agrees with the seeker’s top habitual genre, at one of three intensities (from a mild “maybe” to an insistent request), and the target is replaced by a movie of the requested genre. For *Complement* the requested genre is one that co-occurs with the habitual genres in the catalogue and is not in the tension list $\mathcal{T}$, so it adds a dimension without contradicting the profile. For *Uncertain* the utterance is deliberately vague (“anything is fine, surprise me”), all current-intent cues are removed and the target is left unchanged. Each synthetic instance records its source instance, the injected utterance, the intended relationship and the intensity, and carries an explicit synthetic flag. Synthetic instances appear in the training set as well (with the same flag) so that the relationship head sees clean examples of the rarer classes.

Table II gives the resulting distribution. *Override* is rare in natural data, which is expected: people seldom announce that they are departing from their habits. A slot for human-verified labels exists in the pipeline; no such annotations were available for this run and every table that would have used them reports “not run”.

VI. EXPERIMENTAL SETUP

A. Compared systems

LTP-only and *STI-only* score items from one encoder. *Naive fusion* averages the two encodings with $\alpha = 0.5$; *Adaptive fusion* learns a scalar gate from the two encodings but has no relationship or action head. *Sequential (GRU)* runs a GRU [12]

TABLE II
RELATIONSHIP LABEL COUNTS BY SOURCE AND SPLIT.

Source	Relationship	Train	Test
weak_rule	Complement	1342	165
weak_rule	Conflict	820	48
weak_rule	Consistent	4113	514
weak_rule	Override	83	14
weak_rule	Uncertain	1840	336
synthetic_controlled	Complement	240	31
synthetic_controlled	Conflict	216	33
synthetic_controlled	Consistent	242	24
synthetic_controlled	Override	199	24
synthetic_controlled	Uncertain	247	31

over the concatenation of history items and current-dialogue mentions. *SASRec* [13] stacks two causal self-attention blocks over the same item sequence and scores from the last real position; it does not read dialogue text and is the strongest history-only comparison. *Conversation-aware* attends over the current dialogue with the history vector as query. *KBRD-style* follows the entity-centric design of KBRD [5]: items and genre “entities” mentioned in the dialogue are embedded and pooled with self-attention, fused through a learned gate with the dialogue text encoding, and scored with a genre-aware bias; the external knowledge graph and the response generator of the original are replaced by the shared item tower and MovieLens genres. All four are our own re-implementations in the same parameter budget and item tower; they are not reproductions of any published system and their numbers are not comparable to published ones. The AIPA ablations remove the relationship head, the counterfactual features, clarification, or the persistence tracker, or swap the learned policy for the rule table.

B. Metrics and statistics

a) Ranking: For an instance with target y and fused scores s the rank of the target is the number of catalogue items strictly ahead of it plus one, and the three ranking metrics are functions of that rank alone,

$$r = 1 + |\{i \in \mathcal{I} : s_i > s_y\}|, \quad (32)$$

$$\text{Hit}@K = \mathbb{I}[r \leq K], \quad \text{NDCG}@K = \frac{\mathbb{I}[r \leq K]}{\log_2(r+1)},$$

$$\text{MRR}@K = \frac{\mathbb{I}[r \leq K]}{r}, \quad (33)$$

for $K \in \{10, 20\}$. With one target per instance $\text{Hit}@K$ coincides with $\text{Recall}@K$. Reported values are means over test instances and over the 2 seeds.

b) Conflict subsets: Two natural conflict subsets are reported. The *strict* subset contains the natural test instances whose weak-rule label is Conflict or Override ($n = 62$). The *broad* subset adds instances whose weak label has confidence $\omega_n \geq 0.6$ and whose genre vectors disagree, $\text{JS}(\pi_L, \pi_S) \geq 0.5$ (Jensen–Shannon divergence, base 2), giving $n = 256$. Synthetic Conflict/Override injections form a third subset ($n = 57$) that is never pooled with the natural ones.

c) Relationship classification: With $\hat{r}_n = \arg \max_r p_{n,r}$ we report accuracy, per-class F1 and

$$\text{macro-F1} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} \text{F1}_r, \quad \text{weighted-F1} = \sum_{r \in \mathcal{R}} \frac{N_r}{N} \text{F1}_r, \quad (34)$$

where F1_r is set to zero for a class that is never predicted and never occurs. Calibration uses the confidence $\kappa_n = \max_r p_{n,r}$ and ten equal-width bins $B_b = \{n : \kappa_n \in ((b-1)/10, b/10]\}$,

$$\text{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{N} \left| \frac{1}{|B_b|} \sum_{n \in B_b} \mathbb{I}[\hat{r}_n = r_n] - \frac{1}{|B_b|} \sum_{n \in B_b} \kappa_n \right|, \quad (35)$$

$$\text{Brier} = \frac{1}{|\mathcal{R}^+|} \sum_{r \in \mathcal{R}^+} \frac{1}{N} \sum_{n=1}^N (p_{n,r} - \mathbb{I}[r_n = r])^2, \quad (36)$$

where empty bins are skipped and $\mathcal{R}^+$ is the set of classes that occur at least once in the evaluated subset (one-versus-rest Brier score averaged over those classes) [18].

d) Arbitration and clarification: Let $\hat{a}_n$ be the emitted action, a_n the reference action, and define the per-instance events

$$\begin{aligned} \text{asked}_n &= \mathbb{I}[\hat{a}_n = \text{Ask_Clarification}], \\ \text{conf}_n &= \mathbb{I}[r_n \in \{\text{Conflict}, \text{Override}\}], \\ \text{nec}_n &= \mathbb{I}[r_n = \text{Uncertain}] \vee (\text{conf}_n \wedge \omega_n < \theta_c), \end{aligned} \quad (37)$$

with $\theta_c = 0.6$: a question is deemed necessary when the reference relationship is Uncertain or when it is a conflict that the labelling rule itself was not confident about. Then

$$\begin{aligned} \text{clar. rate} &= \frac{1}{N} \sum_n \text{asked}_n, \\ \text{clar. precision} &= \frac{\sum_n \text{asked}_n \text{nec}_n}{\sum_n \text{asked}_n}, \\ \text{clar. efficiency} &= \frac{\sum_n \text{asked}_n \text{nec}_n}{\sum_n \text{nec}_n}, \\ \text{unnecessary rate} &= \frac{1}{N} \sum_n \text{asked}_n (1 - \text{nec}_n), \end{aligned} \quad (38)$$

and the wrong-override rate is $\frac{1}{N} \sum_n \mathbb{I}[\hat{a}_n = \text{Prioritise_STI} \wedge r_n = \text{Consistent}]$. Arbitration accuracy is $\frac{1}{N} \sum_n \mathbb{I}[\hat{a}_n = a_n]$; conflict-resolution accuracy is the same quantity restricted to $\text{conf}_n = 1$; override success is the mean $\text{Hit}@10$ over instances with $r_n = \text{Override}$ and $\hat{a}_n = \text{Prioritise_STI}$. Precision is undefined when no question is asked.

e) Uncertainty and paired comparisons: Each metric x_n is first averaged over seeds for every test instance n . A 95% interval for a mean is the percentile bootstrap over instances with $B = 200$ resamples,

$$\bar{x}^{(b)} = \frac{1}{N} \sum_{n=1}^N x_{\xi_n^{(b)}}, \quad \xi_n^{(b)} \stackrel{\text{iid}}{\sim} \text{Unif}\{1, \dots, N\}, \quad (39)$$

and the interval is $[\bar{x}_{2.5\%}^{(\cdot)}, \bar{x}_{97.5\%}^{(\cdot)}]$, the empirical percentiles of the resampled means, with a fixed resampling seed. AIPA (full) is compared with every other system on the same instances through the paired differences $d_n = x_n^{\text{AIPA}} - x_n^{\text{other}}$ using the paired t -test and the Wilcoxon signed-rank test (zero differences split between the signs). The m p -values of one

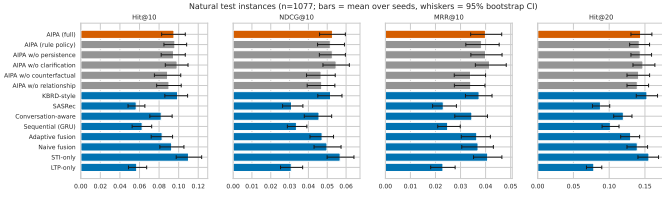


Fig. 3. Hit@10 and NDCG@10 on the natural test set with bootstrap intervals.

family (one metric, one subset) are adjusted by the Holm step-down procedure [19]: with $p_{(1)} \leq \dots \leq p_{(m)}$,

$$\tilde{p}_{(j)} = \max_{l \leq j} \min(1, (m - l + 1) p_{(l)}). \quad (40)$$

Effect sizes are Cohen’s d on the paired differences and Cliff’s δ [20] over all instance pairs,

$$d = \frac{\bar{d}}{s_d}, \quad \delta = \frac{1}{N^2} \sum_{i,j=1}^N (\mathbb{I}[x_i > x'_j] - \mathbb{I}[x_i < x'_j]), \quad (41)$$

where s_d is the sample standard deviation of the d_n , $x = x^{\text{AIPA}}$ and $x' = x^{\text{other}}$, and for $N > 2000$ the sum over j runs over the first 2000 control instances. A comparison whose differences are identically zero (e.g. an ablation that changed no prediction) is reported as undefined rather than as $p = 1$. All natural-data and synthetic-data results are reported separately.

C. Compute

Everything runs on CPU (8 cores; PyTorch 2.14.0+cpu, Python 3.12.13). The configuration used here (“quick” mode) trains for at most 15 epochs with hidden size 64; the released configuration also defines a “full” mode with the complete corpus, more seeds and more bootstrap resamples, which we have not run and for which we make no claims.

VII. RESULTS

A. Overall ranking quality

Table III and Fig. 3 report the natural test set. Two things stand out. First, the intent signal dominates on ReDial: STI-only (0.110) nearly doubles LTP-only (0.057), the two history-only sequential baselines (GRU 0.063, SASRec 0.057) are no better than the profile, and every model that reads the dialogue lands between 0.082 and 0.110 Hit@10. Second, AIPA (full) at 0.095 sits in the upper half of that band, below STI-only and level with the KBRD-style baseline (0.099). It is significantly better than LTP-only ($\Delta = +0.038$, Holm-corrected $p < 0.001$, $d = 0.14$), SASRec ($+0.039$, $p < 0.001$) and the GRU baseline ($+0.032$, $p < 0.001$); its differences from naive fusion, adaptive fusion, the conversation-aware and the KBRD-style baseline are within noise; and it is slightly but significantly *worse* than STI-only ($\Delta = -0.015$, $p = 0.040$, $d = -0.06$; Table IV). The pre-registered success criterion H1 (beat the best baseline on natural Hit@10) is therefore not met.

TABLE III
NATURAL TEST SET ($n = 1077$, MEAN OVER 2 SEEDS; BRACKETS ARE 95% BOOTSTRAP INTERVALS). ROWS ABOVE THE RULE ARE BASELINES.

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.057 [0.049, 0.067]	0.031	0.023	0.078	0.036
STI-only	0.110 [0.097, 0.124]	0.057	0.041	0.156	0.068
Naive fusion	0.093 [0.081, 0.105]	0.050	0.037	0.139	0.062
Adaptive fusion	0.083 [0.072, 0.094]	0.047	0.036	0.130	0.059
Sequential (GRU)	0.063 [0.052, 0.072]	0.034	0.025	0.102	0.043
Conversation-aware	0.082 [0.071, 0.093]	0.046	0.035	0.120	0.055
SASRec	0.057 [0.048, 0.065]	0.031	0.023	0.088	0.039
KBRD-style	0.099 [0.086, 0.109]	0.052	0.037	0.153	0.065
AIPA w/o relationship	0.090 [0.078, 0.103]	0.047	0.034	0.139	0.059
AIPA w/o counterfactual	0.089 [0.075, 0.102]	0.047	0.034	0.141	0.060
AIPA w/o clarification	0.098 [0.086, 0.110]	0.055	0.042	0.147	0.067
AIPA w/o persistence	0.095 [0.082, 0.107]	0.053	0.040	0.143	0.065
AIPA (rule policy)	0.096 [0.085, 0.108]	0.052	0.038	0.142	0.063
AIPA (full)	0.095 [0.083, 0.107]	0.053	0.040	0.144	0.065

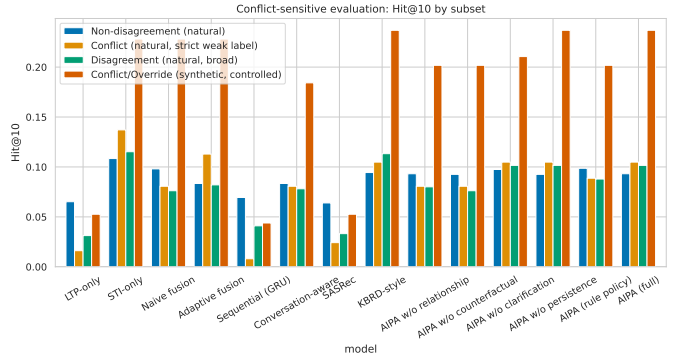


Fig. 4. Hit@10 on natural conflict versus non-conflict turns. Intervals on the conflict side are wide because the subset is small.

B. Conflict turns

The hypothesis that motivates the work is about the turns where LTP and STI disagree. Table V shows the strict natural conflict subset ($n = 62$ under weak-rule labels), the broad subset ($n = 256$) and the controlled synthetic conflict subset ($n = 57$), and Fig. 4 contrasts conflict with non-conflict turns.

On strict natural conflicts STI-only scores highest (0.137), followed by adaptive fusion (0.113); AIPA (full) reaches 0.105, the same as the KBRD-style baseline and above naive fusion (0.081) and the ablations without relationship or counterfactual features (0.081). The confidence intervals are wide (62 instances, two seeds) and no pairwise difference survives correction except against LTP-only and the GRU, so the honest reading is “no evidence either way” among the intent-reading models, with the point estimate leaning towards STI-only. The broad subset (256 turns) tells the same story with tighter intervals: STI-only 0.115, KBRD-style 0.113, AIPA (full) 0.102, adaptive fusion 0.082. On synthetic conflicts every model that trusts STI does well (0.18–0.24), LTP-only, GRU and SASRec collapse to about 0.05, and AIPA (full) ties the KBRD-style baseline at the top (0.237) with STI-only at 0.228, none of the differences significant. The synthetic set therefore confirms that AIPA learns to follow a clearly stated departure, but it also shows that “always follow the intent” is an equally good policy on this data. Criteria H2 (strict and broad) and H3 are not met.

Figure 5 breaks the natural set down by all five relationship classes. The *Uncertain* class (336 instances, a third of them cold seekers) is where the intent-only model gains most of

TABLE IV

PAIRED COMPARISONS OF AIPA (FULL) AGAINST EACH OTHER SYSTEM ON HIT@10. Δ IS THE MEAN PER-INSTANCE DIFFERENCE (POSITIVE FAVOURS AIPA), p_{HOLM} THE HOLM-CORRECTED PAIRED t -TEST p -VALUE (*: < 0.05), d COHEN'S d . DASHES MARK COMPARISONS WITH ZERO VARIANCE (THE PERSISTENCE ABLATION CHANGED NO TEST PREDICTION).

Control	Natural (all, $n=2154$)			Natural conflict ($n=124$)			Synthetic conflict ($n=114$)		
	Δ	p_{Holm}	d	Δ	p_{Holm}	d	Δ	p_{Holm}	d
LTP-only	+0.038	$<0.001^*$	+0.14	+0.089	0.024*	+0.28	+0.184	$<0.001^*$	+0.43
STI-only	-0.015	0.040*	-0.06	-0.032	1.000	-0.10	+0.009	1.000	+0.02
Naive fusion	+0.002	1.000	+0.01	+0.024	1.000	+0.09	+0.009	1.000	+0.03
Adaptive fusion	+0.012	0.184	+0.05	-0.008	1.000	-0.03	+0.009	1.000	+0.02
Sequential (GRU)	+0.032	$<0.001^*$	+0.12	+0.097	0.005*	+0.33	+0.193	$<0.001^*$	+0.44
Conversation-aware	+0.013	0.184	+0.05	+0.024	1.000	+0.09	+0.053	1.000	+0.13
SASRec	+0.039	$<0.001^*$	+0.14	+0.081	0.077	+0.25	+0.184	$<0.001^*$	+0.41
KBRD-style	-0.004	1.000	-0.01	+0.000	1.000	+0.00	+0.000	1.000	+0.00
AIPA w/o relationship	+0.005	1.000	+0.02	+0.024	1.000	+0.08	+0.035	1.000	+0.08
AIPA w/o counterfactual	+0.006	1.000	+0.03	+0.024	1.000	+0.08	+0.035	1.000	+0.08
AIPA w/o clarification	-0.003	0.365	-0.04	—	—	—	+0.026	0.833	+0.16
AIPA w/o persistence	+0.000	1.000	+0.02	—	—	—	—	—	—
AIPA (rule policy)	-0.001	1.000	-0.00	+0.016	1.000	+0.06	+0.035	1.000	+0.09



Fig. 5. Hit@10 per relationship class on natural (weak-rule) and synthetic labels.

its edge; *Consistent* turns, the majority, show all fused models within a few thousandths of each other. Split by history length (Fig. 10), AIPA (full) is competitive on cold seekers (0.132 versus 0.141 for STI-only) and on 10–24 history items (0.106, equal to STI-only), and falls behind on the 766 long-history turns (0.091 versus 0.107), where a profile that is trusted too much appears to cost more than it adds.

C. Relationship classification and arbitration

Table VI reports the relationship head. On natural data accuracy is 0.76 and macro-F1 0.54–0.56 (AIPA (full): 0.540, which meets the pre-set criterion of 0.5); the head is good at *Consistent* and *Uncertain* (F1 0.81 and 0.95), mediocre at *Override* (0.43–0.47, up from about 0.4 before conflict weighting and smoothing were introduced) and *Complement* (0.30–0.43), and poor at *Conflict* (0.15–0.20; criterion of 0.5 not met). Two caveats: the reference labels are themselves weak, so part of this “error” is label noise, and *Conflict* has only 48 natural test instances. On synthetic data the head is near-perfect on *Conflict* and *Override* (F1 0.92 and 0.99) and on the newly injected vague *Uncertain* utterances (1.00), and reaches 0.75 on injected *Complement* requests, which shows it can recognise the patterns when the text states them plainly. The confusion matrix in Fig. 6 shows that natural conflicts are mostly absorbed into *Consistent*.

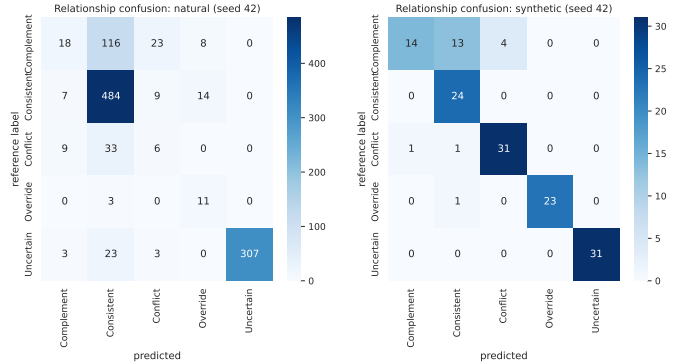


Fig. 6. Relationship confusion matrix of AIPA (full) on natural test turns (rows: weak-rule reference; columns: prediction).

The arbitration head (Table VII, Fig. 7) is the most encouraging part of the results. The learned policy asks a clarifying question on about 20–21% of natural turns, and when it asks, the reference label is *Uncertain* or a low-confidence conflict 97–100% of the time (1.00 for AIPA (full)); unnecessary clarifications are below 1% and wrong overrides below 2%, so both arbitration criteria are met. The rule policy asks far more often (36–40%) with markedly lower precision (0.54–0.62) and an unnecessary-clarification rate of 14–18%, so the learned policy is doing more than reproducing the table it was seeded with. Conflict-resolution accuracy on natural conflicts is low (0.19–0.26 for learned policies, 0.12 for the rule) for the reasons above; on synthetic conflicts it is 0.89–0.93. Calibration of the relationship head is acceptable (ECE 0.07–0.09 on natural data, Fig. 8).

D. Counterfactual driver diagnostic

Table VIII and Fig. 9 summarise the intervention probe for AIPA (full). Removing the STI pathway disrupts the top-10 list far more than removing LTP (overlap with the original list 0.17–0.31 versus 0.57–0.73 on natural turns), and the per-turn

TABLE V
CONFLICT SUBSETS, HIT@10 AND COMPANIONS. NATURAL CONFLICTS USE WEAK-RULE LABELS (STRICT: CONFLICT/OVERRIDE; BROAD: PLUS CONFIDENT GENRE DISAGREEMENT); SYNTHETIC CONFLICTS ARE CONTROLLED INJECTIONS. SAME LAYOUT AS TABLE III.

Natural conflict, strict ($n = 62$)					
Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.016 [0.000, 0.040]	0.005	0.002	0.040	0.010
STI-only	0.137 [0.089, 0.194]	0.058	0.035	0.161	0.064
Naive fusion	0.081 [0.040, 0.129]	0.033	0.019	0.121	0.043
Adaptive fusion	0.113 [0.064, 0.177]	0.055	0.038	0.145	0.063
Sequential (GRU)	0.008 [0.000, 0.024]	0.003	0.001	0.056	0.015
Conversation-aware	0.081 [0.040, 0.129]	0.038	0.026	0.121	0.049
SASRec	0.024 [0.000, 0.056]	0.017	0.014	0.024	0.017
KBRD-style	0.105 [0.056, 0.153]	0.047	0.029	0.202	0.071
AIPA w/o relationship	0.081 [0.040, 0.137]	0.037	0.024	0.137	0.051
AIPA w/o counterfactual	0.081 [0.040, 0.137]	0.036	0.023	0.153	0.054
AIPA w/o clarification	0.105 [0.056, 0.161]	0.050	0.034	0.145	0.060
AIPA w/o persistence	0.105 [0.056, 0.161]	0.046	0.028	0.137	0.053
AIPA (rule policy)	0.089 [0.040, 0.153]	0.048	0.036	0.137	0.060
AIPA (full)	0.105 [0.056, 0.161]	0.046	0.028	0.137	0.053

Natural conflict, broad ($n = 256$)					
Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.031 [0.018, 0.045]	0.015	0.010	0.047	0.018
STI-only	0.115 [0.092, 0.139]	0.057	0.039	0.164	0.069
Naive fusion	0.076 [0.057, 0.098]	0.038	0.027	0.123	0.050
Adaptive fusion	0.082 [0.061, 0.105]	0.042	0.030	0.121	0.052
Sequential (GRU)	0.041 [0.025, 0.061]	0.022	0.016	0.076	0.031
Conversation-aware	0.078 [0.055, 0.100]	0.037	0.025	0.113	0.046
SASRec	0.033 [0.018, 0.051]	0.018	0.013	0.066	0.026
KBRD-style	0.113 [0.086, 0.141]	0.061	0.045	0.170	0.075
AIPA w/o relationship	0.080 [0.057, 0.102]	0.038	0.025	0.148	0.055
AIPA w/o counterfactual	0.076 [0.055, 0.098]	0.036	0.024	0.152	0.055
AIPA w/o clarification	0.102 [0.076, 0.131]	0.053	0.038	0.145	0.064
AIPA w/o persistence	0.102 [0.076, 0.133]	0.053	0.038	0.143	0.063
AIPA (rule policy)	0.088 [0.066, 0.113]	0.047	0.034	0.135	0.058
AIPA (full)	0.102 [0.076, 0.133]	0.053	0.038	0.143	0.063

Synthetic conflict ($n = 57$)					
Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.053 [0.009, 0.096]	0.029	0.021	0.053	0.029
STI-only	0.228 [0.158, 0.316]	0.111	0.075	0.395	0.153
Naive fusion	0.228 [0.166, 0.316]	0.120	0.088	0.404	0.164
Adaptive fusion	0.228 [0.149, 0.307]	0.111	0.076	0.342	0.140
Sequential (GRU)	0.044 [0.009, 0.079]	0.022	0.015	0.070	0.029
Conversation-aware	0.184 [0.123, 0.254]	0.101	0.077	0.254	0.119
SASRec	0.053 [0.018, 0.096]	0.024	0.015	0.079	0.031
KBRD-style	0.237 [0.140, 0.342]	0.130	0.098	0.465	0.187
AIPA w/o relationship	0.202 [0.140, 0.263]	0.100	0.070	0.342	0.136
AIPA w/o counterfactual	0.202 [0.140, 0.263]	0.100	0.071	0.342	0.136
AIPA w/o clarification	0.211 [0.140, 0.289]	0.113	0.082	0.395	0.160
AIPA w/o persistence	0.237 [0.167, 0.316]	0.120	0.084	0.430	0.167
AIPA (rule policy)	0.202 [0.140, 0.272]	0.101	0.071	0.377	0.145
AIPA (full)	0.237 [0.167, 0.316]	0.120	0.084	0.430	0.167

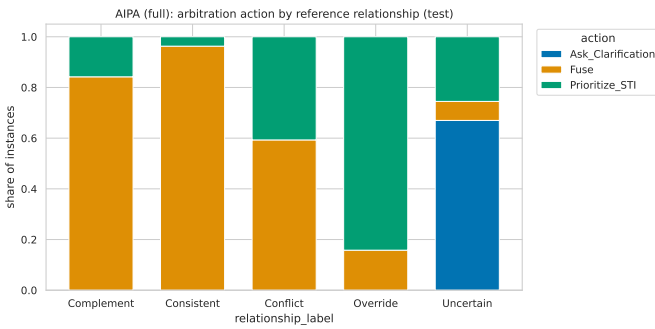


Fig. 7. Distribution of predicted actions within each reference relationship class, AIPA (full), natural turns.

driver label is *STI-driven* or *Jointly-driven* almost everywhere; *LTP-driven* turns are about 1% and no turn is *Neither-driven*. On synthetic conflicts and overrides the STI dependence is strongest (overlap without STI 0.02, 82–85% STI-driven), which is what one would want. Agreement between the driver label and the chosen action is 0.50 on natural turns and 0.55 on synthetic turns. The diagnostic thus confirms what the ranking

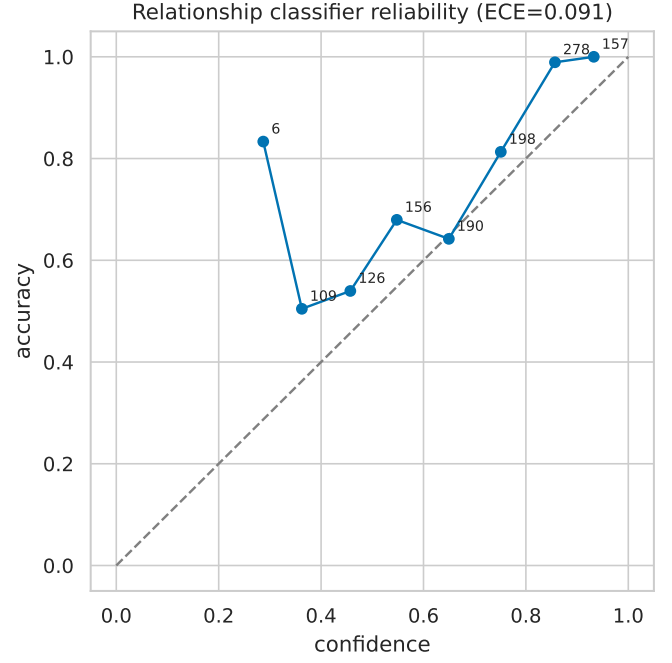


Fig. 8. Reliability diagram of the relationship head.

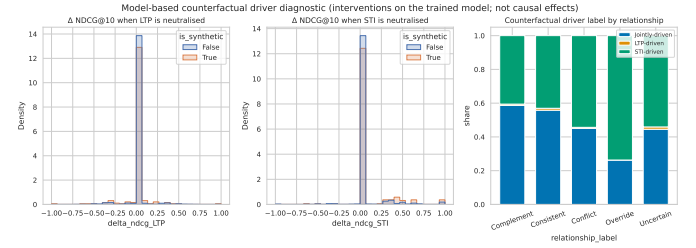


Fig. 9. Driver-label distribution and NDCG changes under signal removal.

tables already suggest: the trained model leans heavily on the conversation and uses the profile as a secondary signal.

E. Persistent shifts

With $k = 2$ the tracker flagged five persistent shifts per seed for AIPA (full), 40 across all AIPA variants and seeds (e.g. seeker 1087 repeatedly steering toward war films, seekers 1008 and 1011 toward horror). Because a shift is only applied to instances that come *after* the second qualifying session, it touched 121 test instances (nine seekers), on which Hit@10 moved from 0.087 without the tracker to 0.091 with it (+0.004, permutation $p = 1$); over the whole natural set the difference is +0.0005. A sweep over $k \in \{1, 2, 3\}$ on the validation split gives 41, 1.5 and 0 shifts respectively and no change in validation Hit@10 at any k . The mechanism works as designed but is not exercised enough by this data budget to be evaluated; the criterion that the tracker improves the instances it affects is not met.

F. Sensitivity and fusion weight

Figure 10 plots Hit@10 against history length, number of seeker turns and injected conflict intensity. Performance

TABLE VI
RELATIONSHIP CLASSIFICATION, MEAN OVER SEEDS. W-F1 IS WEIGHTED F1; THE LAST FIVE COLUMNS ARE PER-CLASS F1 (–: CLASS ABSENT FROM THE SUBSET).

Model	Subset	Acc.	Macro-F1	W-F1	Compl.	Consist.	Confl.	Overr.	Uncert.
AIPA w/o relationship	natural	0.763	0.556	0.758	0.433	0.806	0.162	0.433	0.944
AIPA w/o relationship	synthetic	0.895	0.895	0.893	0.749	0.843	0.903	0.989	0.992
AIPA w/o counterfactual	natural	0.761	0.550	0.754	0.424	0.805	0.151	0.429	0.939
AIPA w/o counterfactual	synthetic	0.885	0.885	0.883	0.729	0.829	0.896	0.979	0.992
AIPA w/o clarification	natural	0.760	0.544	0.745	0.303	0.812	0.186	0.468	0.950
AIPA w/o clarification	synthetic	0.913	0.911	0.910	0.785	0.855	0.924	0.989	1.000
AIPA w/o persistence	natural	0.759	0.538	0.744	0.305	0.810	0.177	0.449	0.951
AIPA w/o persistence	synthetic	0.902	0.899	0.898	0.749	0.842	0.915	0.989	1.000
AIPA (rule policy)	natural	0.760	0.546	0.745	0.322	0.809	0.195	0.459	0.944
AIPA (rule policy)	synthetic	0.909	0.910	0.909	0.795	0.867	0.928	0.967	0.992
AIPA (full)	natural	0.760	0.540	0.744	0.304	0.811	0.177	0.459	0.951
AIPA (full)	synthetic	0.902	0.899	0.898	0.749	0.842	0.915	0.989	1.000

TABLE VII
ARBITRATION AND CLARIFICATION BEHAVIOUR, MEAN OVER SEEDS. CLARIFICATION PRECISION IS UNDEFINED WHEN NO QUESTION IS ASKED.

Model	Subset	Arb. acc.	Confl. res.	Overr. succ.	Clar. rate	Clar. prec.	Unnec. clar.	Wrong overr.
AIPA w/o relationship	natural	0.887	0.185	0.243	0.209	0.971	0.006	0.013
AIPA w/o relationship	synthetic	0.948	0.895	0.304	0.217	1.000	0.000	0.000
AIPA w/o counterfactual	natural	0.887	0.226	0.236	0.211	0.965	0.007	0.013
AIPA w/o counterfactual	synthetic	0.944	0.886	0.310	0.220	0.984	0.003	0.000
AIPA w/o clarification	natural	0.687	0.258	0.263	0.000	–	0.000	0.018
AIPA w/o clarification	synthetic	0.734	0.930	0.208	0.000	–	0.000	0.000
AIPA w/o persistence	natural	0.882	0.218	0.270	0.200	1.000	0.000	0.018
AIPA w/o persistence	synthetic	0.941	0.904	0.229	0.217	1.000	0.000	0.000
AIPA (rule policy)	natural	0.746	0.121	0.633	0.399	0.543	0.184	0.008
AIPA (rule policy)	synthetic	0.832	0.842	0.300	0.360	0.623	0.143	0.000
AIPA (full)	natural	0.882	0.218	0.270	0.200	1.000	0.000	0.019
AIPA (full)	synthetic	0.941	0.904	0.229	0.217	1.000	0.000	0.000

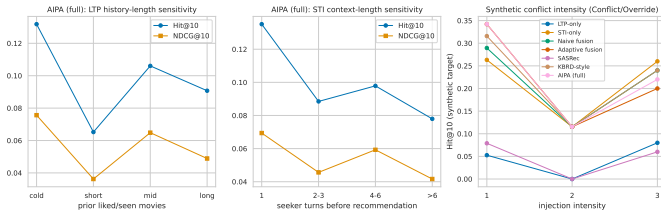


Fig. 10. Sensitivity of Hit@10 to history length, STI length and synthetic conflict intensity.

decreases with longer histories (0.132 for 0–2 history items, 0.065 for 3–9, 0.106 for 10–24 and 0.091 for 25 or more), a hint that long profiles add noise more than signal in the current encoder. On synthetic conflicts, AIPA (full) reaches 0.34 at the mildest intensity (STI-only 0.26), 0.12 at the middle one and 0.22 at the strongest (STI-only 0.26); with 13–25 instances per cell these differences are not interpretable. Table IX shows the fixed-weight sweep for naive fusion: anything between $\alpha_{LTP} = 0.25$ and 0.5 is close to the best (0.092–0.093), and pure LTP weighting collapses to 0.031, which sets the scale for how much the profile alone can achieve here.

G. Ablations and efficiency

Reading the AIPA rows of Tables III and IV as an ablation study, none of the removals changes natural Hit@10 by more than 0.007 and none is significant. Removing the relationship head or the counterfactual features now costs 0.005–0.007 Hit@10 on natural turns and 0.024 on strict conflicts (the sign one would hope for, but far from significant), while removing clarification yields the numerically best natural score (0.098). All variants have 0.48–0.55 M parameters, train in seconds to under a minute on CPU and score a turn in under 0.2 ms (Table X); the arbitration machinery costs roughly 6% extra parameters and a factor of about 2.4 in inference time relative to naive fusion, both negligible in absolute terms.

H. Qualitative cases and error analysis

Table XI shows representative test turns. The pattern is consistent with the aggregate numbers: when a seeker states a genre plainly (“war movies this weekend similar to Dunkirk”), the model reads it as an override, prioritises STI and ranks the target reasonably; when the request is indirect (“good old fashioned monster movies for my brother”), the STI genre cue is empty, the model asks, and the target sits far down the list. The most common error type is a weak-rule *Complement* or *Conflict* turn predicted as *Consistent* and fused; in the natural set 75% of *Complement* and 81% of *Conflict* references are

TABLE VIII

COUNTERFACTUAL PROBE OF AIPA (FULL) BY RELATIONSHIP CLASS: MEAN ABSOLUTE CHANGE IN NDCG@10 AND TOP-10 OVERLAP WHEN ONE PATHWAY IS REMOVED, AND THE SHARE OF TURNS ASSIGNED TO EACH DRIVER LABEL. THESE DESCRIBE THE MODEL, NOT THE USERS.

Source	Relationship	n	$ \Delta\text{NDCG@10} $		Top-10 overlap		Driver (%)			
			no LTP	no STI	no LTP	no STI	STI	LTP	Joint	Neither
natural	Complement	165	0.027	0.056	0.613	0.210	40	1	59	0
natural	Conflict	48	0.024	0.033	0.567	0.191	35	1	64	0
natural	Consistent	514	0.027	0.048	0.609	0.280	43	1	55	0
natural	Override	14	0.043	0.074	0.725	0.168	54	0	46	0
natural	Uncertain	336	0.014	0.050	0.718	0.305	57	1	42	0
synthetic	Complement	31	0.089	0.091	0.621	0.200	42	0	58	0
synthetic	Conflict	33	0.021	0.127	0.848	0.020	82	0	18	0
synthetic	Consistent	24	0.020	0.027	0.579	0.315	35	0	65	0
synthetic	Override	24	0.039	0.122	0.875	0.017	85	0	15	0
synthetic	Uncertain	31	0.028	0.070	0.569	0.487	27	2	71	0

TABLE IX

FIXED FUSION WEIGHT SWEEP (NAIVE FUSION, NATURAL TEST SET UNLESS STATED).

α_{LTP}	Hit@10	NDCG@10	Hit@20	Hit@10 (synthetic)
0.00	0.084	0.047	0.131	0.178
0.25	0.092	0.050	0.137	0.175
0.50	0.093	0.050	0.139	0.168
0.75	0.073	0.040	0.126	0.122
1.00	0.031	0.016	0.055	0.017

TABLE X

MODEL SIZE AND COST (CPU ONLY).

Model	Parameters	Size (MB)	Train (s)	CPU inf. (ms/sample)
LTP-only	515,605	2.06	3.7	0.032
STI-only	515,605	2.06	6.0	0.047
Naive fusion	515,605	2.06	3.6	0.031
Adaptive fusion	526,870	2.11	14.8	0.062
Sequential (GRU)	492,877	1.97	19.1	0.048
Conversation-aware	476,173	1.91	4.9	0.013
SASRec	526,413	2.11	55.6	0.176
KBRD-style	477,814	1.91	4.3	0.021
AIPA w/o relationship	547,230	2.19	11.0	0.071
AIPA w/o counterfactual	547,230	2.19	14.9	0.068
AIPA w/o clarification	547,550	2.19	24.1	0.096
AIPA w/o persistence	547,550	2.19	12.3	0.073
AIPA (rule policy)	535,386	2.14	14.4	0.085
AIPA (full)	547,550	2.19	11.9	0.074

mispredicted, and their targets have median ranks of 427 and 348.

VIII. DISCUSSION

A. What the evidence supports

Four things are supported by this run. (i) Any model that reads the conversation beats a profile-only model on ReDial by a wide, significant margin; the profile alone, whether pooled (LTP-only) or modelled sequentially (GRU, SASRec), reaches about 6% Hit@10. (ii) The learned arbitration policy is well-behaved: it asks when it should, rarely asks when it should not, and rarely overrides in the wrong direction. (iii) On plainly stated departures from habit (the synthetic set), AIPA follows the user, and the counterfactual probe shows that the ranking really is driven by the stated intent in those cases. (iv) Making the fusion table learnable and training harder on

disagreement turns (conflict weighting, label smoothing, self-training, all selected on validation) moved AIPA from below to level with the strongest fusion baselines on conflict turns and raised Override-F1, without changing the overall picture.

B. What it does not support

The central claim, that explicit arbitration improves recommendations over simpler fusion *specifically on conflict turns*, is not supported here. The strict natural conflict subset is too small for a decisive test and its point estimate is below STI-only; the broad subset agrees; on synthetic conflicts AIPA merely ties the KBRD-style baseline and the trivial intent-following policy. On the full natural set the intent-only model is in fact slightly and significantly better. The ablations are flat. We see three reasons, in rough order of importance.

First, the labels. Weak-rule relationship labels are the reference for the relationship head, the arbitration metrics and the conflict subset itself. If the rule mislabels many turns, we are training toward noise and evaluating on a subset that is not what we think it is. The near-perfect synthetic results versus the weak natural results point in that direction.

Second, the budget. A quarter of the corpus and two seeds give 62 strict (256 broad) natural conflict turns. With Hit@10 around 0.1 that is nowhere near enough to detect a plausible effect of a few points.

Third, the representation. TF-IDF/SVD text features capture genre words and little else. Much of what makes a request an intent (“for my brother”, “something to fall asleep to”) is invisible to the STI encoder, and so the *Uncertain* class is large and cold-start heavy.

C. What a conclusive test needs

- 1) Run the “full” configuration: the whole corpus, at least five seeds and 1,000 bootstrap resamples, so that the conflict subset has several hundred turns.
- 2) Obtain human relationship annotations for a few hundred test turns. The pipeline already accepts such a file and would report all metrics against it separately; this is the single most valuable addition.
- 3) Run with the pre-trained sentence encoder (MiniLM) that the “full” configuration already selects for both

TABLE XI
SELECTED TEST TURNS (N: NATURAL, S: SYNTHETIC). REFERENCE LABELS ARE WEAK-RULE LABELS; “RANK” IS THE RANK OF THE TRUE TARGET UNDER AIPA (FULL).

	Dialogue excerpt (seeker turns before the target)	LTP profile	STI signal	Ref./pred. relationship	Action	w_{LTP}/w_{STI}	Driver	Rank
N	Recommender: Yes great decade for actions! Seeker: Oh course! :) Seeker: I really liked True Lies (1994) and Speed (1994) Seeker: What do you think?	Drama 0.24; Comedy 0.17; Romance 0.16	Action 0.63; Romance 0.13; Thriller 0.13	Complement / Consistent	Fuse	0.49/0.51	STI	961
N	Recommender: Hi Recommender: What kind of movies do you like? Seeker: Hiya. I like rom-com movies. Can you recommend any?	Drama 0.24; Thriller 0.16; Mystery 0.11	Comedy 0.50; Romance 0.50	Consistent / Consistent	Fuse	0.39/0.61	Jointly	13
N	Seeker: Yes, I loved it Seeker: I love movies with Will Ferrell too, like Anchorman Recommender: What about Jumanji (2017) Seeker: Oh, I haven't seen Jumanji (...)	Crime 0.38; Drama 0.38; Film-Noir 0.09	Comedy 0.58; Adventure 0.08; Children 0.08	Conflict / Consistent	Fuse	0.41/0.59	Jointly	334
N	Recommender: hi there! I would like to recommend some movies to ya. What kind of movies do you like? Seeker: I;d like to see some war movies this weekend similar to ...	Comedy 0.26; Drama 0.22; Romance 0.13	War 0.68; Drama 0.18; Romance 0.06	Override / Override	Prioritize_STI	0.22/0.78	STI	2120
N	Recommender: Hello! How are you Seeker: I'm good! My brother is coming to visit and I want to queue up some good old fashioned monster movies for him. Any suggesti...	Comedy 0.28; Horror 0.18; Sci-Fi 0.17	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.54/0.46	Jointly	4789
N	Recommender: maybe you feel like an action movie today? Die Hard (1988) or The Transporter (2002) Seeker: Yes I have actually. Can you suggest any other action comed...	Action 0.31; Crime 0.17; Thriller 0.13	Action 0.42; Comedy 0.42; Crime 0.08	Consistent / Consistent	Fuse	0.47/0.53	Jointly	54
N	Seeker: American Recommender: i could recommend you too American Pie 2 (2001) Recommender: There are many of those movies Seeker: Im not sure if i saw american p...	(none: cold seeker)	Comedy 1.00	Uncertain / Uncertain	Prioritize_STI	0.22/0.78	STI	1227
N	Seeker: haa ok ... that if I'm interested .. all my friends saw it and they tell me it's the maximum Seeker: I'll look for her this weekend Recommender: It (2017)...	Action 0.36; Sci-Fi 0.22; Drama 0.15	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.50/0.50	STI	1295

seeker utterances and item descriptions; the run reported here uses TF-IDF/SVD.

- 4) Compare against a faithful knowledge-grounded baseline (KBRD/KGSF with their knowledge graph), since the KBRD-style re-implementation here already matches AIPA and a positive result would have to hold against the original models.
- 5) Revisit the labels for *Conflict*: with the loss weights, conflict weighting and smoothing now tuned on validation, Conflict-F1 is still below 0.2 against the weak rule, which points at the reference rather than at the optimiser.
- 6) Evaluate clarification with users, or at least with simulated answers; offline precision says nothing about whether a question was welcome.

D. Threats to validity

Conversation identifiers are assumed chronological; ReDial seekers are crowd workers, not organic users, so cross-session “preferences” may partly reflect task instructions; the MovieLens join misses a fraction of titles, leaving those items without genres; and the sequential, SASRec, conversation-aware and KBRD-style baselines are our own small re-implementations. All of these are documented in the released code and reported in the generated experiment report.

IX. CONCLUSION

We set out to make the reconciliation of long-term preference and short-term intent explicit in a conversational recommender and to test whether doing so helps where it should,

on turns where the two disagree. We built a complete, reproducible pipeline on an English corpus, with leak-free cross-session histories, two clearly separated label sources covering all five relationship classes, a relationship classifier trained with conflict-weighted and self-trained weak labels, a learned arbitration policy with learned fusion weights and clarification, a persistence tracker and a counterfactual probe, and we evaluated it against eight baselines with intervals, corrected paired tests and pre-stated success criteria. The reduced-budget results show a policy that behaves sensibly and a model that follows clearly stated intent and matches a KBRD-style baseline, but they do not show a ranking advantage over simpler fusion or over an intent-only model, and we have not hidden that. The next step is not another model component but better labels and a larger run; the code is ready for both.

REFERENCES

- [1] K. Christakopoulou, F. Radlinski, and K. Hofmann, “Towards conversational recommender systems,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 815–824.
- [2] Y. Sun and Y. Zhang, “Conversational recommender system,” in *Proceedings of the 41st International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2018, pp. 235–244.
- [3] W. Lei, X. He, Y. Miao, Q. Wu, R. Hong, M.-Y. Kan, and T.-S. Chua, “Estimation-action-reflection: Towards deep interaction between conversational and recommender systems,” in *Proceedings of the 13th International Conference on Web Search and Data Mining (WSDM)*, 2020, pp. 304–312.
- [4] Y. Zhang, X. Chen, Q. Ai, L. Yang, and W. B. Croft, “Towards conversational search and recommendation: System ask, user respond,” in *Proceedings of the 27th ACM International Conference on Information and Knowledge Management (CIKM)*, 2018, pp. 177–186.

- [5] Q. Chen, J. Lin, Y. Zhang, M. Ding, Y. Cen, H. Yang, and J. Tang, "Towards knowledge-based recommender dialog system," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 1803–1813.
- [6] K. Zhou, W. X. Zhao, S. Bian, Y. Zhou, J.-R. Wen, and J. Yu, "Improving conversational recommender systems via knowledge graph based semantic fusion," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2020, pp. 1006–1014.
- [7] R. Li, S. E. Kahou, H. Schulz, V. Michalski, L. Charlin, and C. Pal, "Towards deep conversational recommendations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.
- [8] S. A. Hayati, D. Kang, Q. Zhu, W. Shi, and Z. Yu, "INSPIRED: Toward sociable recommendation dialog systems," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 8142–8152.
- [9] K. Zhou, Y. Zhou, W. X. Zhao, X. Wang, and J.-R. Wen, "Towards topic-guided conversational recommender system," in *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020, pp. 4128–4139.
- [10] D. Jannach, A. Manzoor, W. Cai, and L. Chen, "A survey on conversational recommender systems," *ACM Computing Surveys*, vol. 54, no. 5, pp. 1–36, 2021.
- [11] C. Gao, W. Lei, X. He, M. de Rijke, and T.-S. Chua, "Advances and challenges in conversational recommender systems: A survey," *AI Open*, vol. 2, pp. 100–126, 2021.
- [12] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, "Session-based recommendations with recurrent neural networks," in *International Conference on Learning Representations (ICLR)*, 2016.
- [13] W.-C. Kang and J. McAuley, "Self-attentive sequential recommendation," in *2018 IEEE International Conference on Data Mining (ICDM)*, 2018, pp. 197–206.
- [14] M. Quadrana, P. Cremonesi, and D. Jannach, "Sequence-aware recommender systems," *ACM Computing Surveys*, vol. 51, no. 4, pp. 1–36, 2018.
- [15] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press, 2009.
- [16] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.
- [17] F. M. Harper and J. A. Konstan, "The MovieLens datasets: History and context," *ACM Transactions on Interactive Intelligent Systems*, vol. 5, no. 4, pp. 1–19, 2015.
- [18] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [19] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [20] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychological Bulletin*, vol. 114, no. 3, pp. 494–509, 1993.